Lab 17 Worksheet — Power Supply Sizing, Connectors and Safety

Name: ____ **Date:** __

Exam objective: Select a power supply by wattage, form factor, efficiency and connector complement, and apply electrical safety rules when working with power (Core 1 objectives 3.5 and 4.5).

Record as you go

Step	What you did	What you observed
1	Record the absolute safety rule first: never open a power supply unit. Its capacitors hold a lethal charge long after it is unplugged, and there are no user-serviceable parts inside.	
2	Record the second safety rule: disconnect the mains lead and hold the power button for 15 seconds before working inside any machine.	
3	Build the power budget by listing each component with its typical and peak draw: CPU, GPU, motherboard, each drive, each fan and any PCIe card.	
4	Sum the peak figures to get total system draw, then add 30 percent headroom for capacitor ageing, transient spikes and future upgrades.	
5	Record the connector complement your build requires: 24-pin ATX for the board, 4 or 8-pin EPS for the CPU, 6 or 8-pin PCIe for the graphics card, SATA power for each drive and Molex for legacy devices.	
6	Match the form factor to the case: ATX for standard and micro-ATX cases, SFX for Mini-ITX and small-form-factor builds.	
7	Compare the 80 PLUS efficiency tiers from Bronze through Titanium, and explain that a	

Step	What you did	What you observed
	higher tier wastes less power as heat and runs cooler and quieter.	
8	Explain the difference between a modular, semi-modular and non-modular supply, and why modular improves airflow in a small case.	
9	Record the standard voltage rails — plus 12 V, plus 5 V and plus 3.3 V — and note that the 12 V rail carries the CPU and GPU load and is the one that matters most.	
10	Observe a demonstration of testing a supply with a multimeter or a PSU tester, and record the acceptable tolerance of plus or minus 5 percent on each rail.	
11	List the symptoms of a failing power supply: no power at all, random shutdowns under load, a burning smell, a fan that does not spin and repeated POST failures.	
12	Write the final selection with its wattage, form factor, efficiency tier and connector list, and justify the wattage against your calculated budget plus headroom.	

Verification

Success criterion: Your power budget sums every component and adds stated headroom, the selected supply provides every connector the build requires, and the safety procedure explicitly forbids opening the PSU and requires discharge before internal work.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
